

Aim

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LK1 re-sequenced

Code ▼

260522_VH01624_461_222JLJVN

2026-08-03

Run: 260522_VH01624_461_222JLJVN (re-sequencing of 260423_VH01624_453_222HWMYX)

Status: current. This is the LK1 data used by all later analyses; the pilot run 260423_VH01624_453_222HWMYX it replaces is superseded.

Analysis finished: 2026-06-01 - the most recent of the 13 figures shown below.

Report generated: 2026-08-03 by 05.LK1_resequenced_report.Rmd , from the outputs of steps 01-04 in this folder.

LK1 was sequenced a second time to decide whether the pilot depth was sufficient, and whether the rest of the cohort needed deeper sequencing. The comparison is against the pilot run and against the published AML scAtlas.

For full report, download LK1_resequenced_report.html in the attachment and open in a browser.

Code used - attached and in the github repository:

https://github.com/nbartonicek/surface_proteome_dev/tree/main/lab_archives/LK1-resequenced
(https://github.com/nbartonicek/surface_proteome_dev/tree/main/lab_archives/LK1-resequenced)

Aim

Identify whether we are getting a comparable amount of key features to the AML scAtlas - cells, genes and UMIs - and evaluate whether further sequencing is required, via:

- subsampling of reads
- differential expression between cell types
- separation of clusters

Conclusions

- AML scAtlas basic QC stats are comparable to our dataset, on average lower
- subsampling of reads shows that we have reached saturation levels for cell detection
- we have **not** reached saturation levels for: number of genes detected; number of UMIs per gene; number of DE genes for cells with either low (T-cells) or high (HSCs) numbers of genes expressed
- cluster separation of HSCs from other precursors (MEP) is slightly better for re-sequenced data
- re-sequencing gained about 25% more genes and 39% more UMIs per cell for essentially no extra cells, which is the expected shape when a library is under-sequenced rather than under-loaded

1. What re-sequencing gained

First run vs re-sequenced, per-cell gain

name	first_run	resequenced	absolute_gain	fold_gain
median_genes	2052.000	2562.000	510.0000	1.248538
median_UMI	4202.000	5833.000	1631.0000	1.388149
mean_genes	2595.986	3095.429	499.4427	1.192390
mean_UMI	8396.377	11630.430	3234.0527	1.385172

Our two runs against the AML scAtlas

dataset	cells	median_genes	mean_genes	p10_genes	p90_genes	median_UMI	mean_UMI	p10_UMI	p90_UMI
AML scAtlas	748679	1611	1801.281	690	3228	2195.882	2158.776	1448.527	2829.167
first_run	23133	2052	2595.986	920	5539	4202.000	8396.377	1479.000	24312.400
resequenced	23569	2562	3095.429	1115	6393	5833.000	11630.430	1892.800	34087.000

Cell numbers barely move while genes and UMIs per cell rise by a quarter and a third respectively. That is the signature of a library that was sequenced too shallowly rather than loaded too lightly - more reads recover more of the transcriptome from the cells already captured, rather than finding new cells.

Both runs sit **above** the atlas medians on genes and UMIs per cell, which is what “atlas stats are on average lower” in the conclusions refers to.

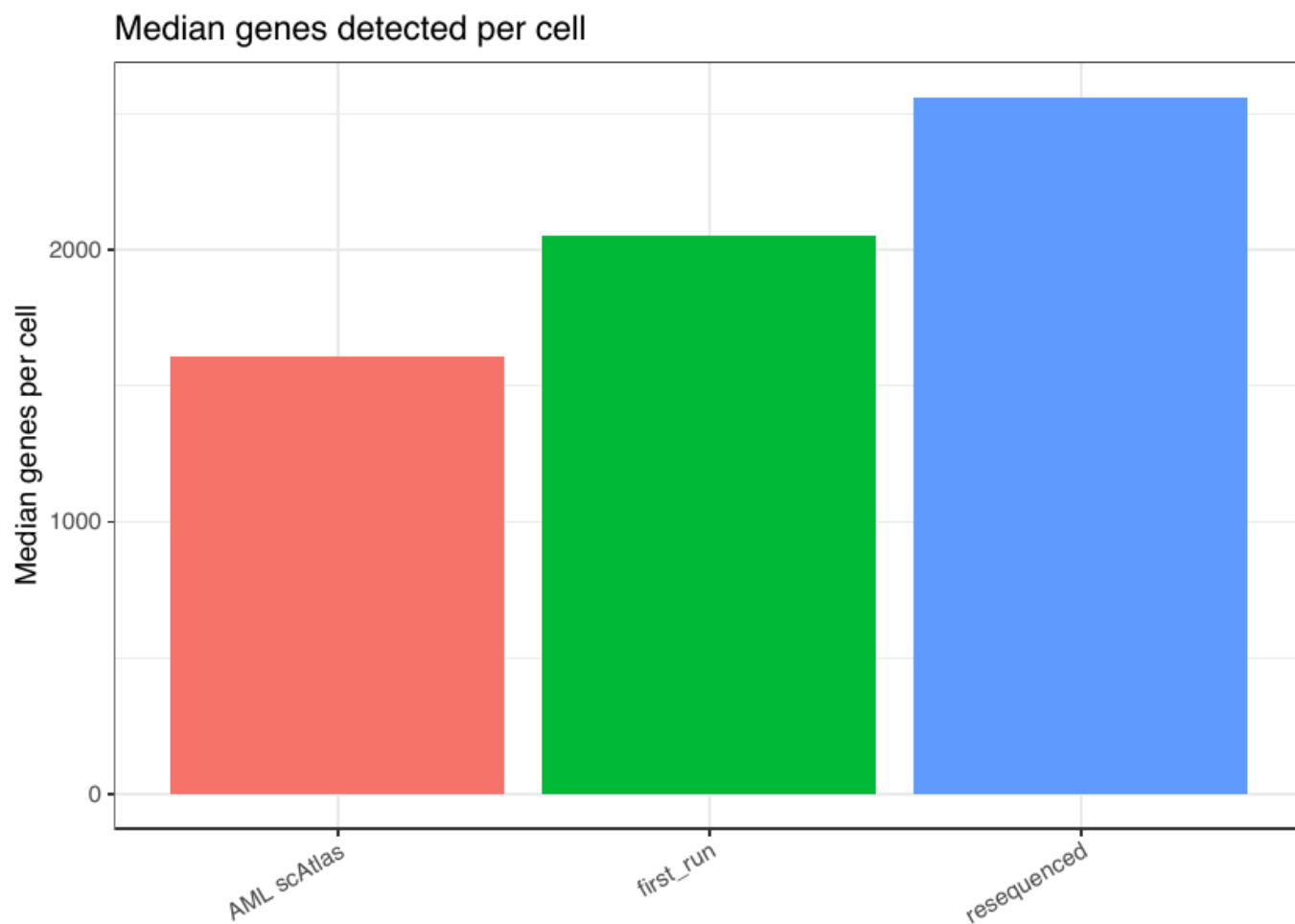
Cell Ranger metrics, both runs

Estimated Number of Cells	Mean Reads per Cell	Median Genes per Cell	Number of Reads	Sequencing Saturation
23569	29125	2562	686439168	46.5%
23133	20537	2052	475071355	45.0%

Sequencing saturation stays in the mid-40s in both runs - consistent with the curves below, where cell detection has plateaued but per-cell content has not.

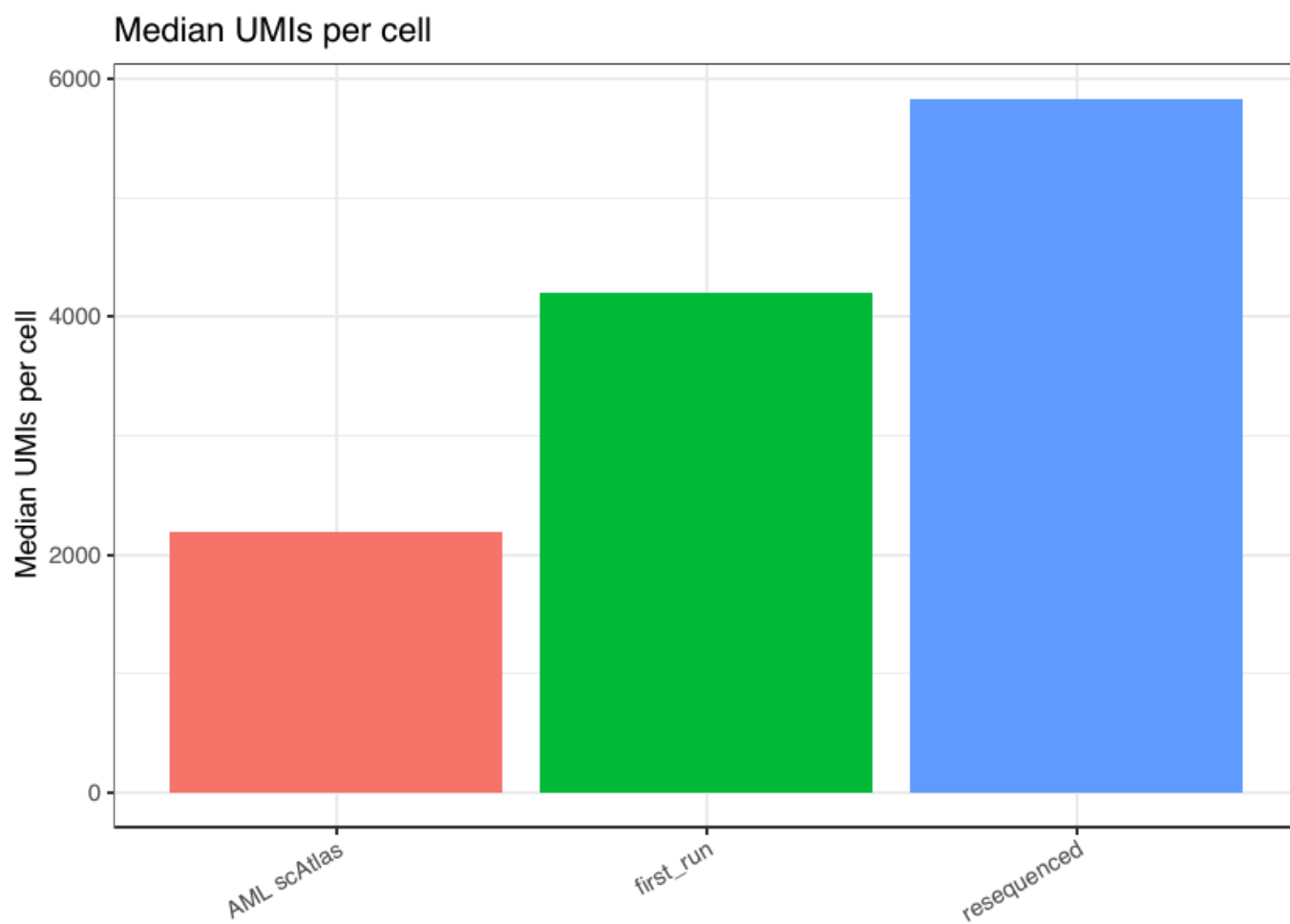
median_genes_barplot.pdf

Median genes per cell, first run vs re-sequenced.



median_UMIs_barplot.pdf

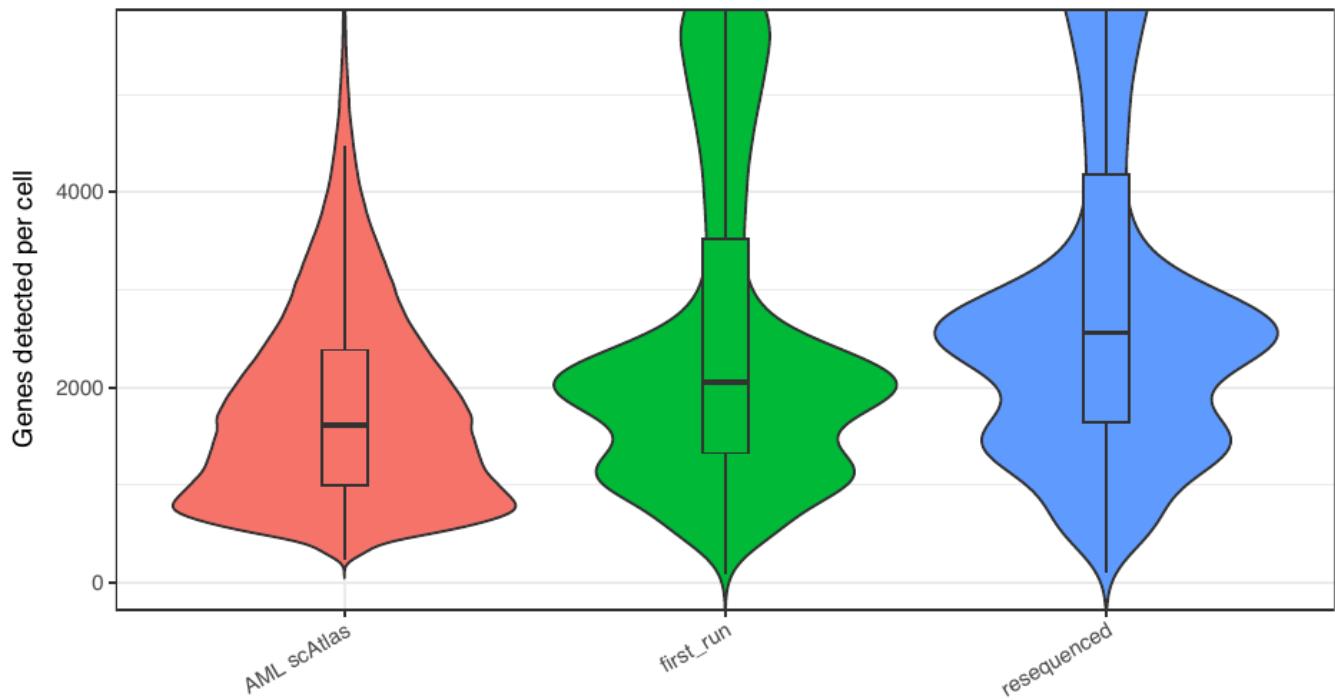
Median UMIs per cell, first run vs re-sequenced.



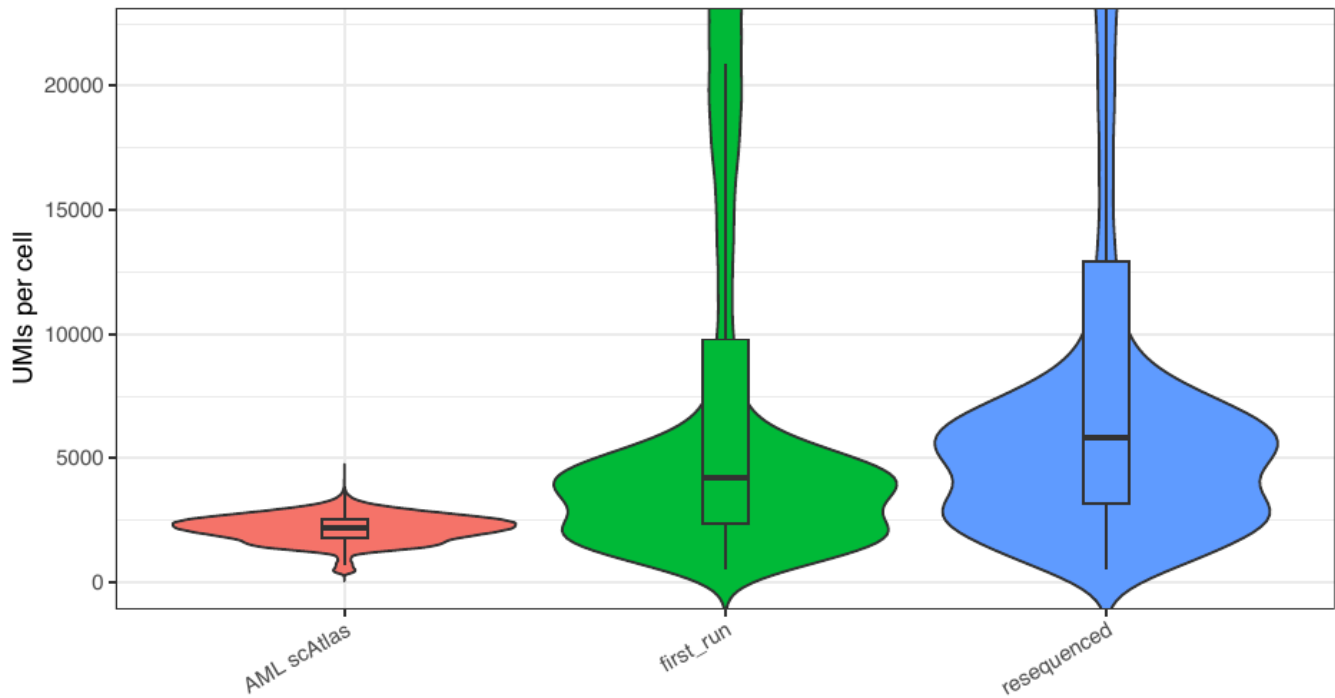
combined_benchmark_plot.pdf

Combined benchmark across both runs and the atlas.

Genes per cell: our Cell Ranger outputs vs AML scAtlas



UMIs per cell: our Cell Ranger outputs vs AML scAtlas

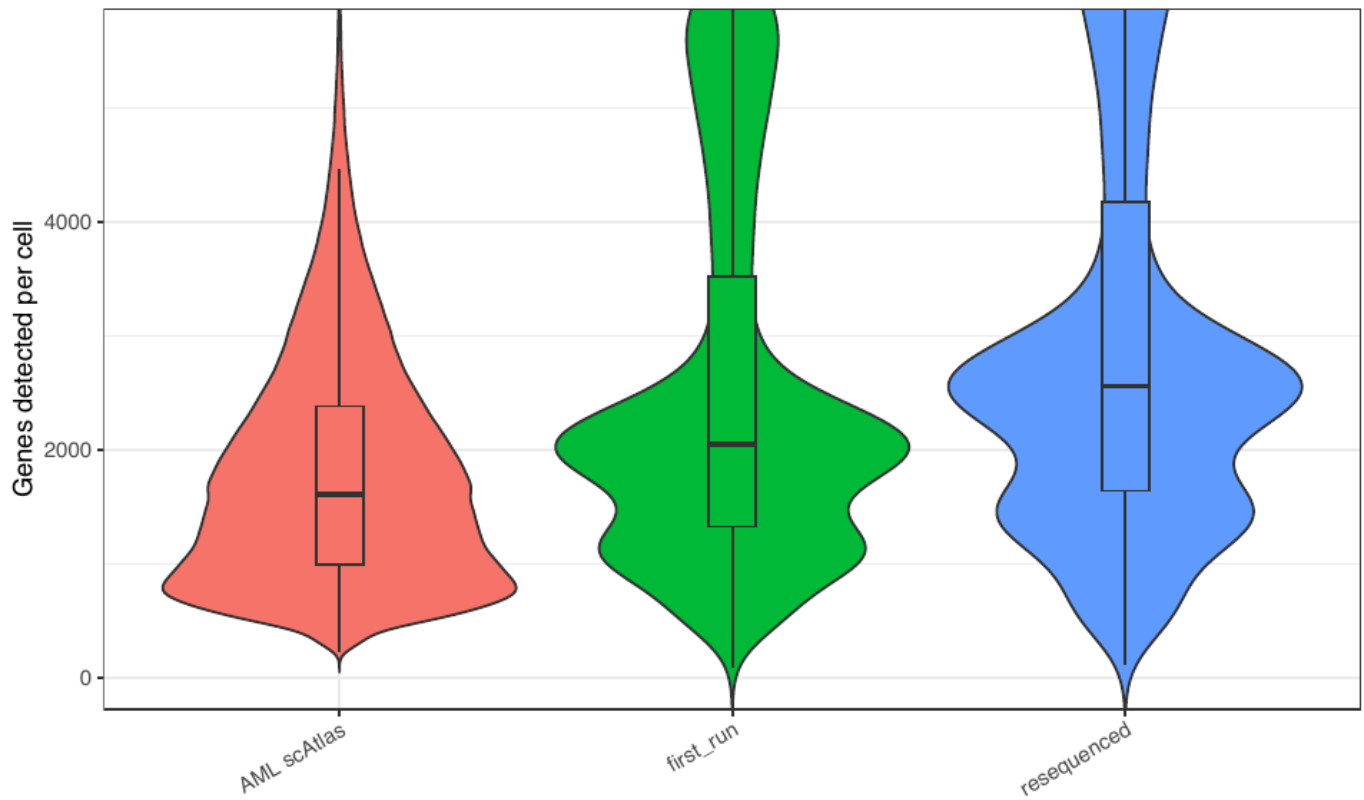


2. Against the AML scAtlas

`comparison_genes_vs_AML_atlas.pdf`

Genes per cell, our runs against the atlas.

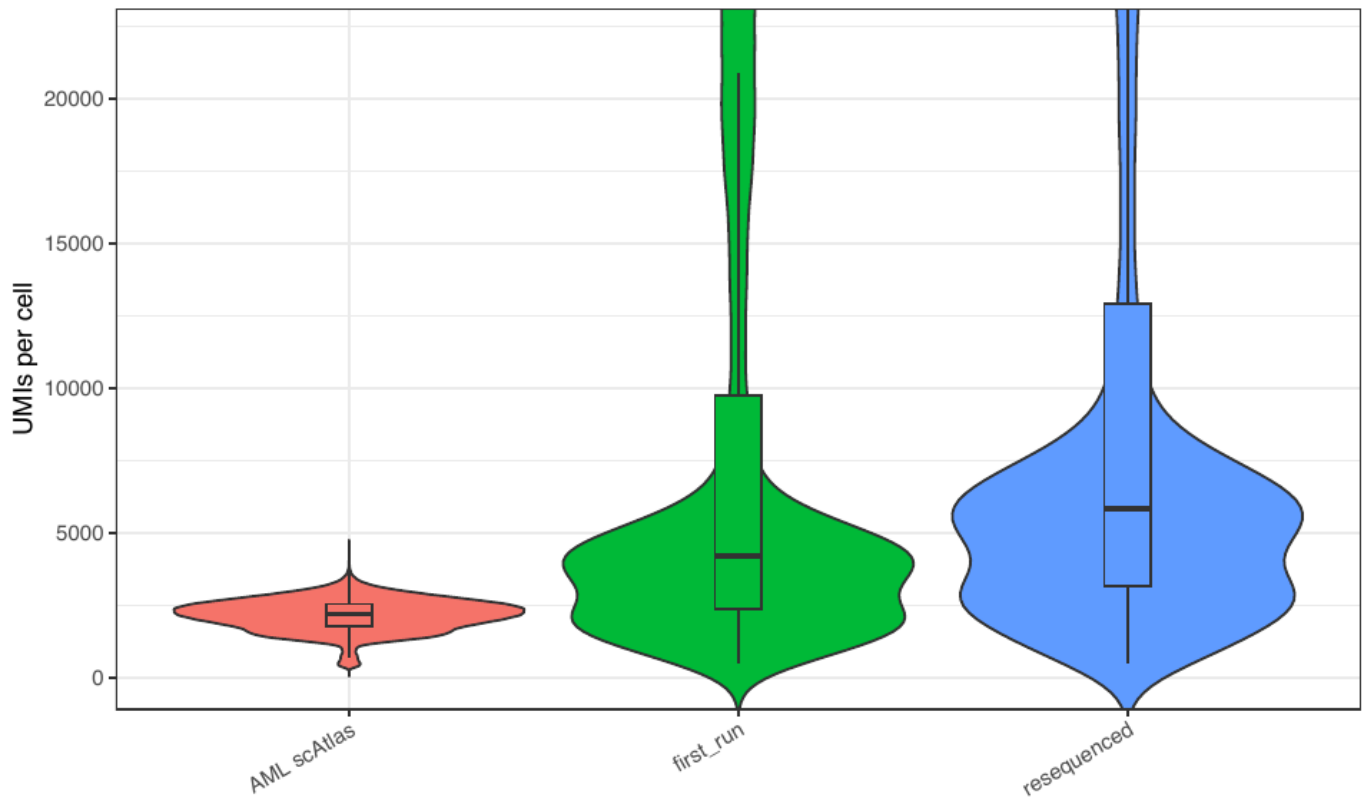
Genes per cell: our Cell Ranger outputs vs AML scAtlas



[comparison_UMIs_vs_AML_atlas.pdf](#)

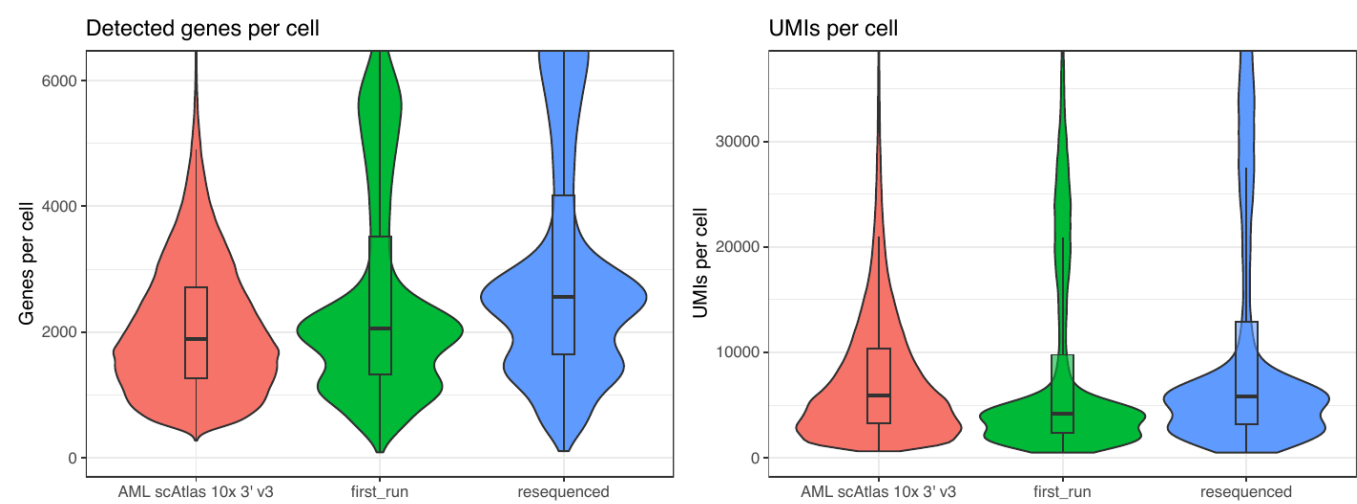
UMIs per cell, our runs against the atlas.

UMIs per cell: our Cell Ranger outputs vs AML scAtlas



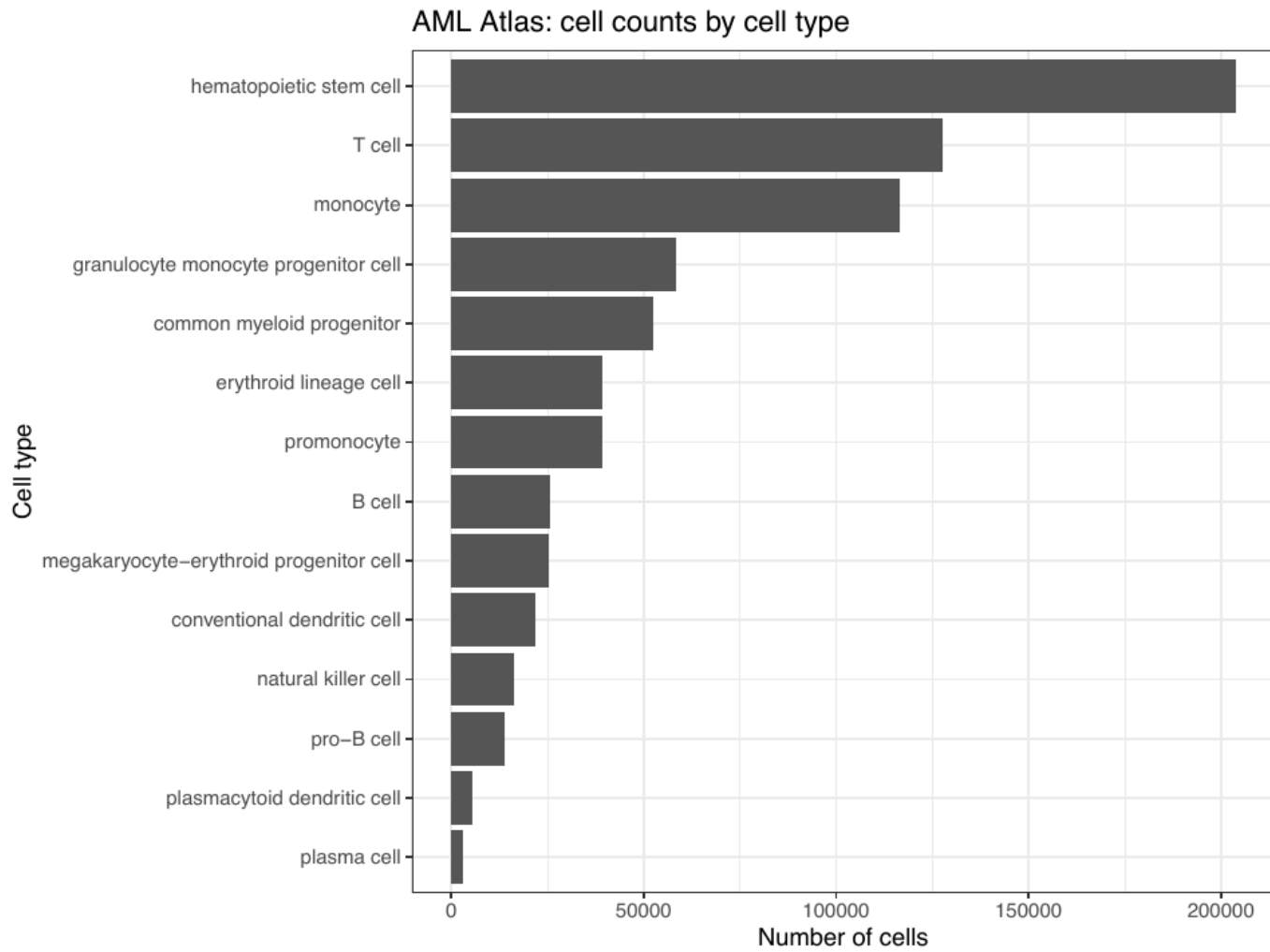
[violin_genes_UMIs_per_cell_full_depth_vs_AML_10x3v3.pdf](#)

Restricted to 10X 3' v3, the assay that matches ours - the fair comparison, since the atlas pools several chemistries with different capture efficiency.



barplot_cell_counts_by_cell_type.pdf

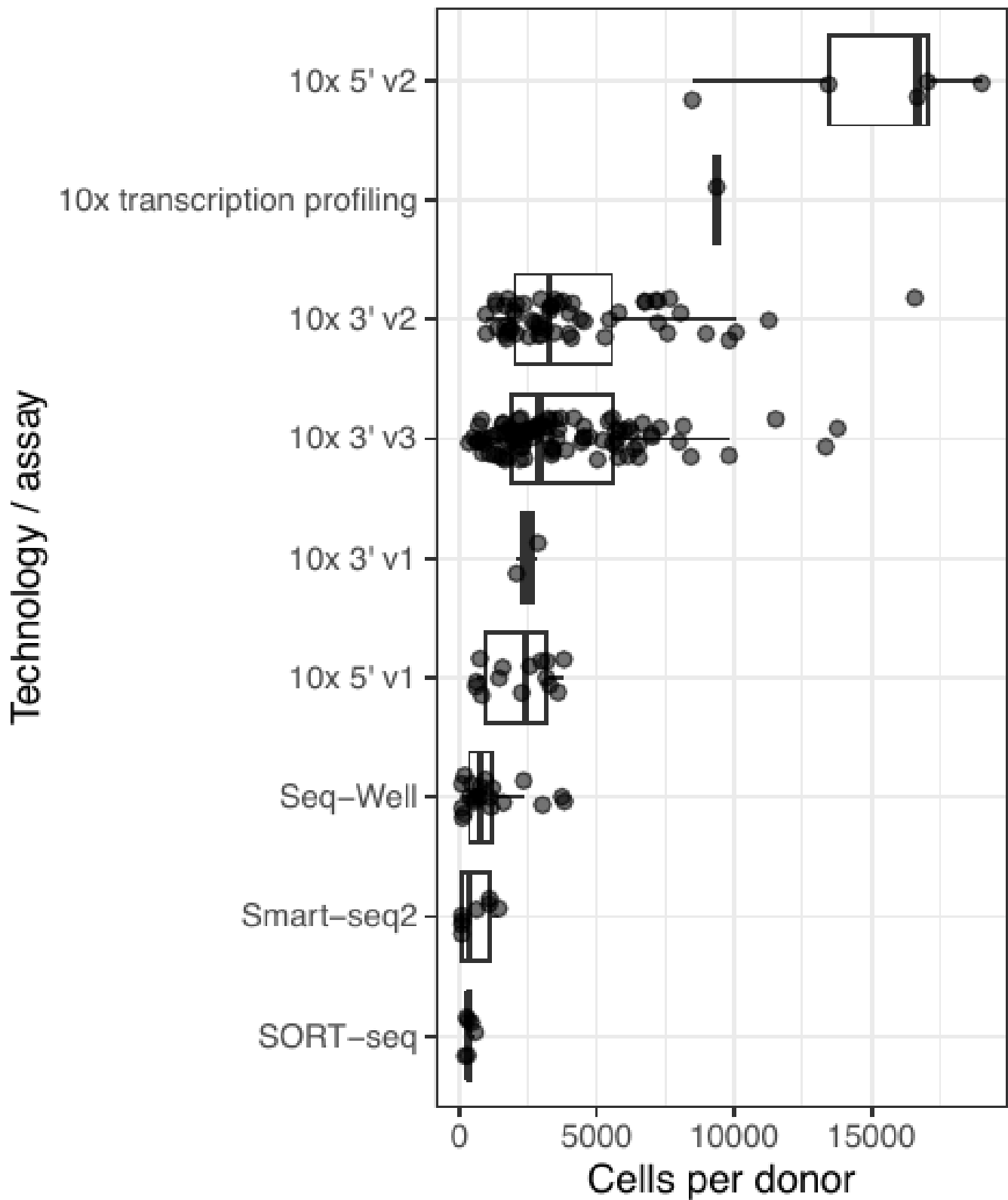
Atlas cell counts by cell type, for context on what the reference contains.



boxplot_cells_per_donor_by_assay.pdf

Cells per donor by assay in the atlas.

Cells per donor by assay

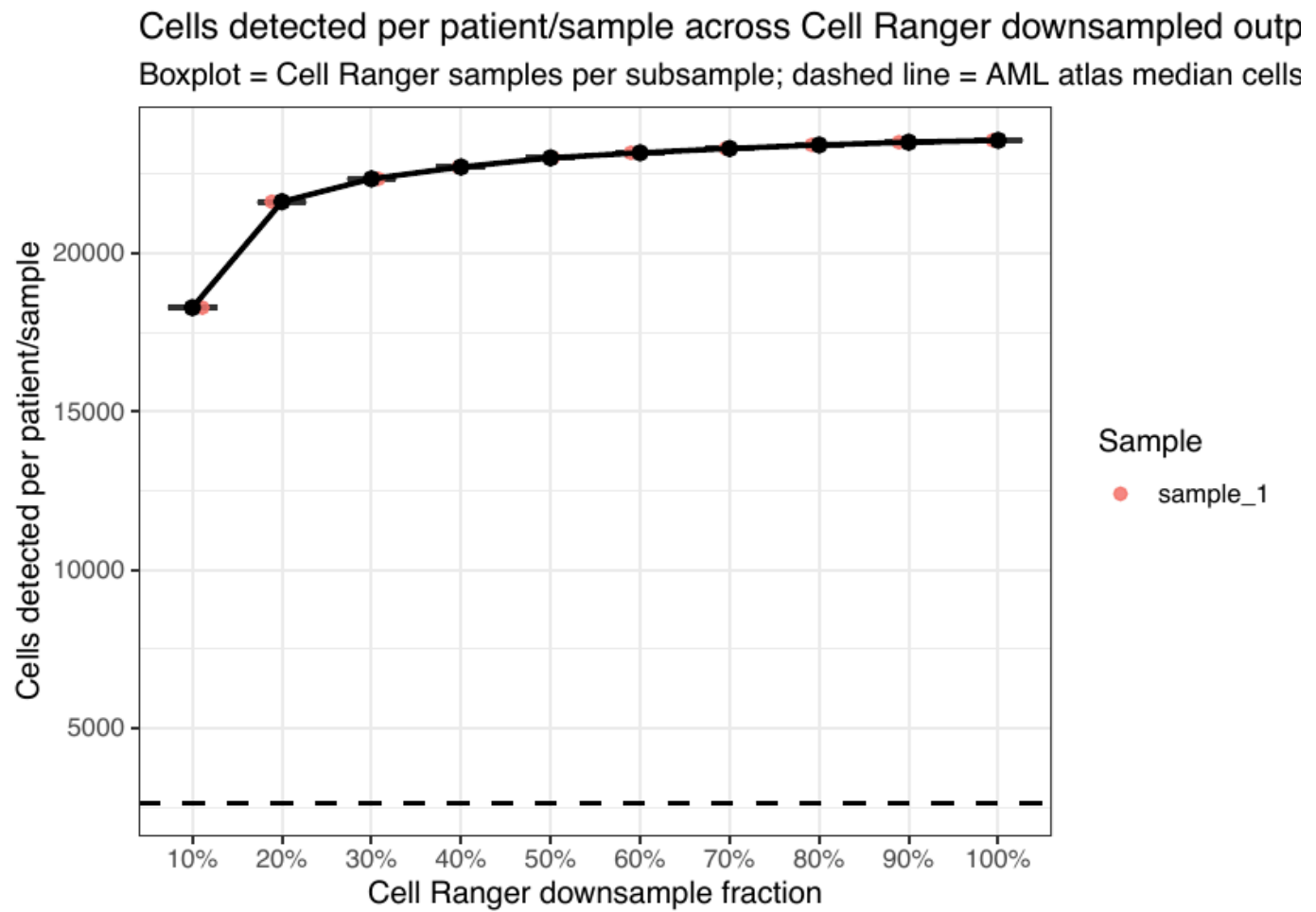


3. Saturation curves

Cell Ranger's molecule-level downsampling, re-analysed at each depth and plotted against the atlas. These are the figures the saturation conclusions rest on.

[curve_cells_detected_vs_AML.pdf](#)

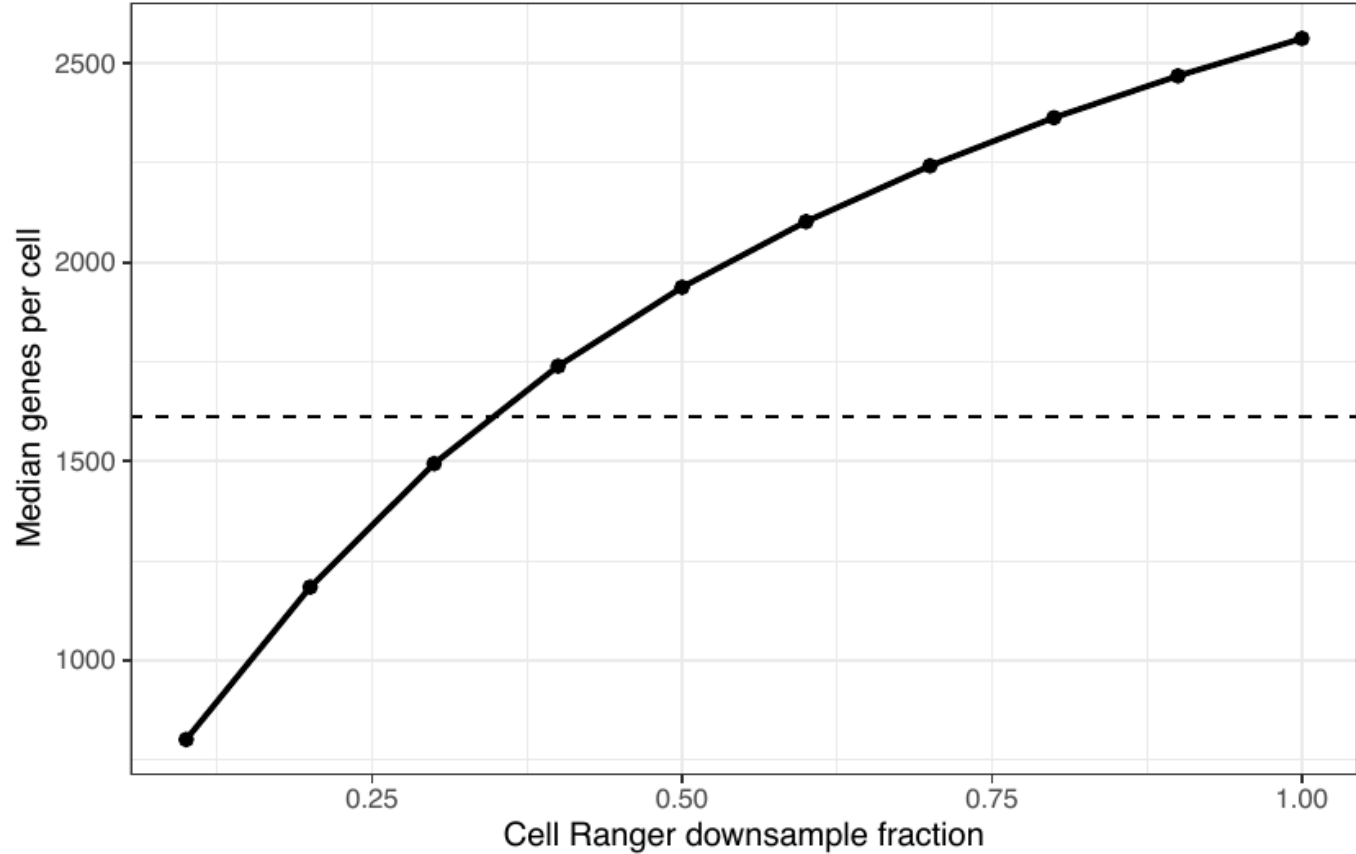
Cells detected vs depth - this one has plateaued. Extra reads do not recover extra cells.



curve_genes_per_cell_vs_AML.pdf

Genes per cell vs depth - still climbing. No plateau, so more depth would still add genes.

Genes per cell across Cell Ranger downsampled outputs
Dashed line = AML atlas reference

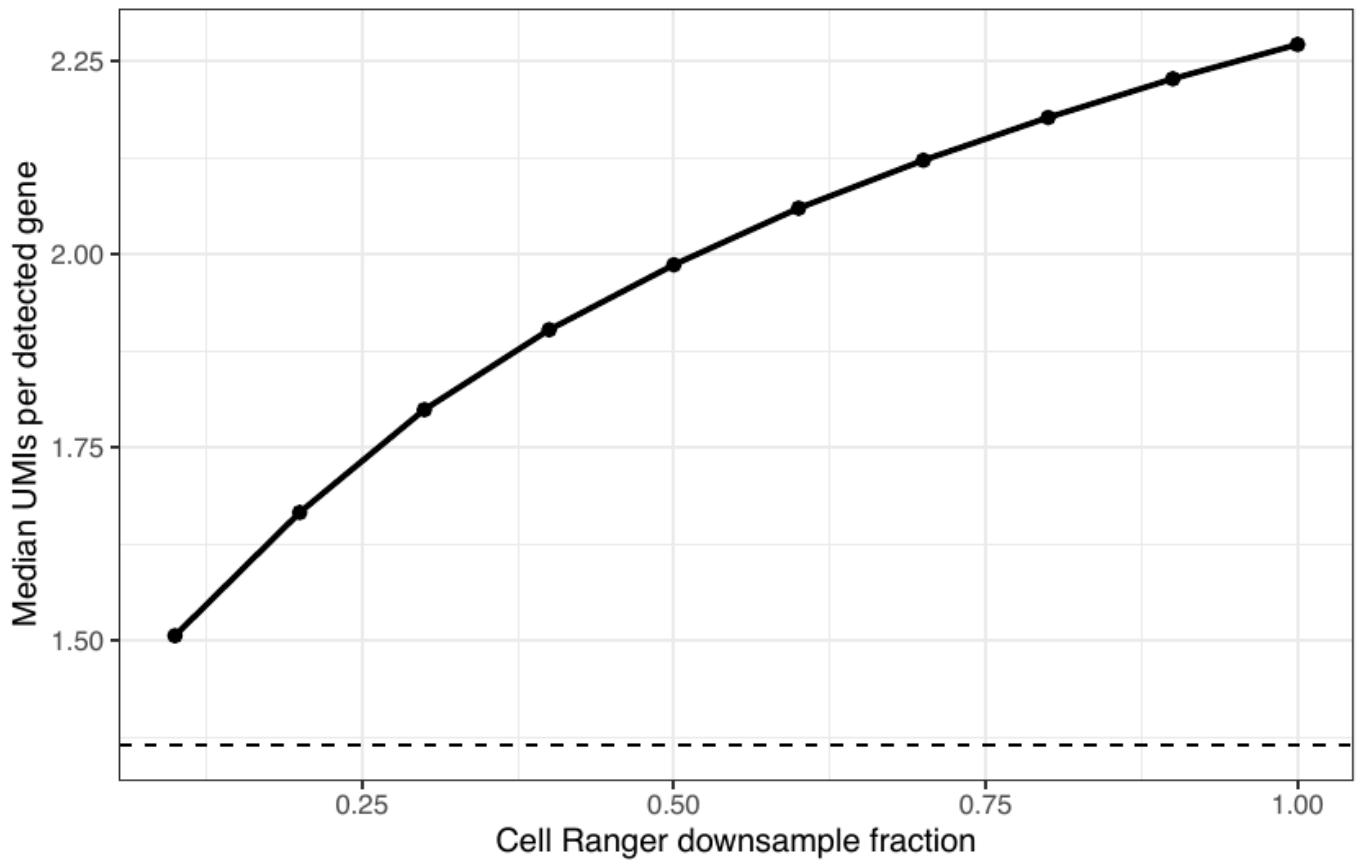


curve_UMIs_per_gene_vs_AML.pdf

UMIs per gene vs depth - still climbing.

UMIs per gene across Cell Ranger downsampled outputs

Dashed line = AML atlas reference

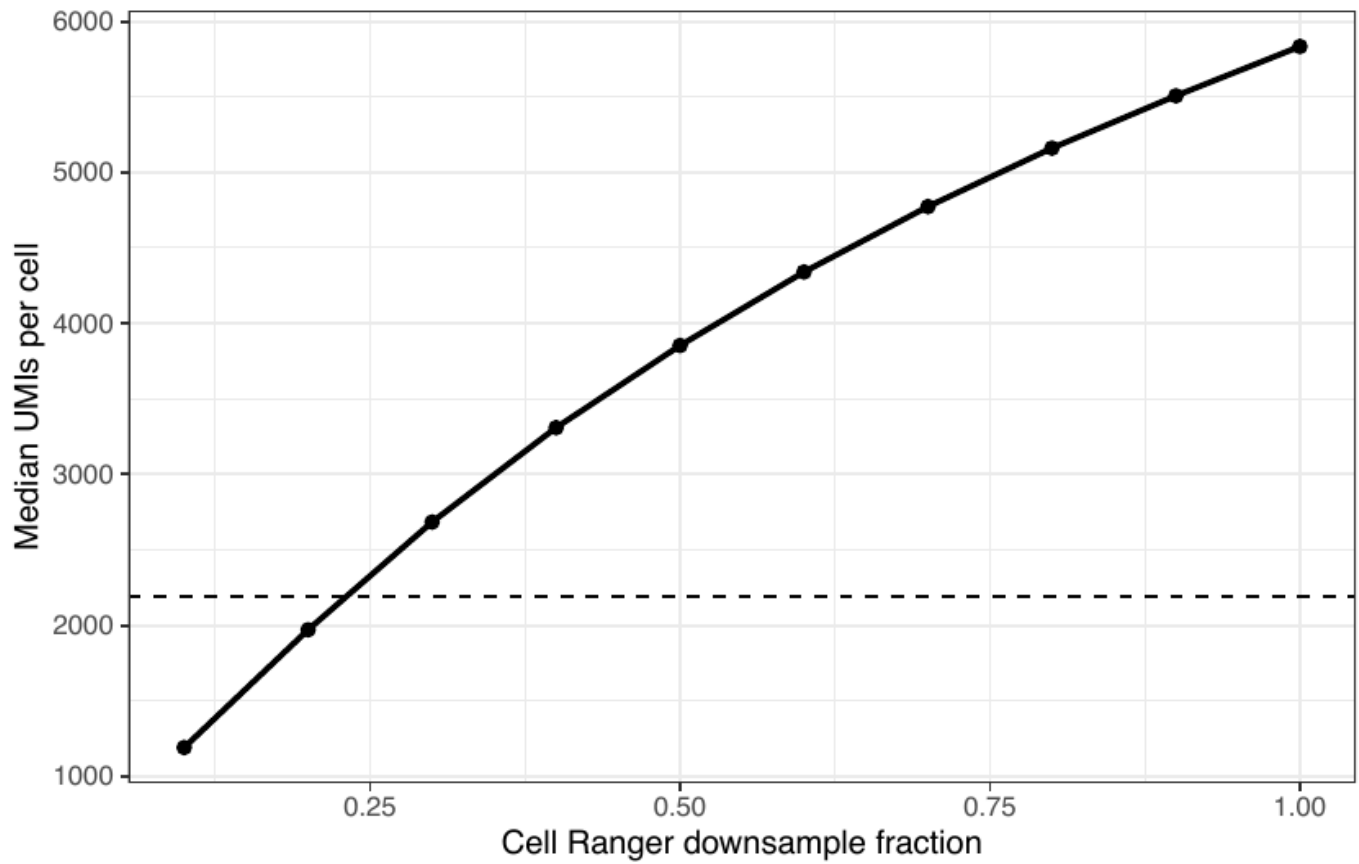


curve_UMIs_per_cell_vs_AML.pdf

UMIs per cell vs depth.

UMIs per cell across Cell Ranger downsampled outputs

Dashed line = AML atlas reference

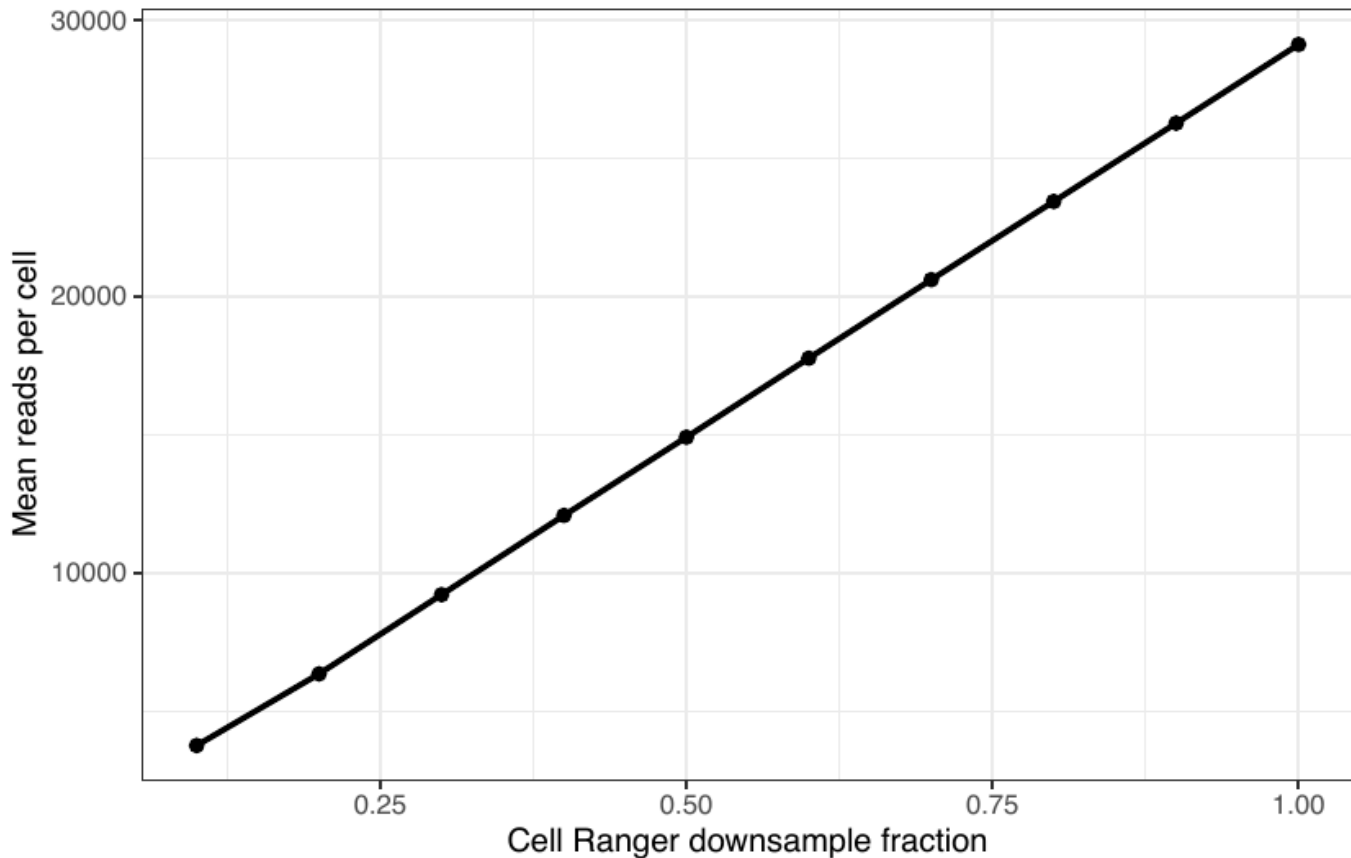


`curve_reads_per_cell.pdf`

Reads per cell across the depth series.

Reads per cell across Cell Ranger downsampled outputs

Reads per cell unavailable for AML atlas h5ad count matrix



4. Effect of depth on downstream analysis

Two things were checked beyond raw counts: whether differential expression between cell types holds up at lower depth, and whether clusters still separate.

Cluster counts across the depth series, from the analysis in step 07 (reproduced from the read-depth deck, as the figure itself is not among the saved outputs):

Depth	Total clusters	HSC clusters
10% subsample	5	2
30% subsample	8	3
60% subsample (~pilot 1)	8	4
Full dataset	9	4

The full dataset resolves one cluster the 60% subsample does not - naive T-cells - and HSC separation from other precursors improves with depth. The pilot run sits at roughly the 60% point of the re-sequenced data, which is why its HSC/MEP separation is the weaker of the two.

5. Data locations

What	Location
Sequencing facility output	/pipeline/Runs/NextSeq/260522_VH01624_461_222JLJVN/ProjectFolders/Project_Lev-Kats

What	Location
Working copy, cluster	/scratch/users/nbartonicek/projects/amgen/raw_reseq/260522_VH01624_461_222JLJVN - Sample_LK1-GEX/
Cell Ranger output	/scratch/users/nbartonicek/projects/amgen/results/cellranger_withbam/260522_VH01624_461_222JLJVN
Depth benchmarking	results/benchmarking/depth_benchmarking/ and ../depth_benchmarking_cellranger_downsampled/
Reference	Cell Ranger annotation refdata-gex-GRCh38-2024-A

Table: Output directories under results/

/260522_VH01624_461_222JLJVN (top-level entries)

directory	entries	first	last
cellranger_withbam	1	2026-06-25	2026-06-25
cite_seq_count	3	2026-06-03	2026-07-08
emptydrops	2	2026-06-02	2026-06-02
genotype_demux	2	2026-06-02	2026-06-02
scDblFinder	1	2026-06-02	2026-06-02
seurat_demux	6	2026-06-02	2026-06-09
cellbender	1	2026-06-24	2026-06-24

The re-sequenced run was then taken through the standard processing - emptyDrops, CITE-seq-Count, cellSNP/vireo, scDblFinder, CellBender. Those steps are benchmarked in their own reports rather than here; this one covers the re-sequencing decision only.

6. What was run, in order

Step	Script	What it does
01	01.cellranger_resequence.sh	Cell Ranger 10.0.0 count on the re-sequenced LK1 GEX library.
02	02.investigate_resequencing.R	First run vs re-sequenced: cells, genes, UMIs, and the gain summary.
03	03.subsampling_vs_atlas.R	Our depth against the AML scAtlas, plus the molecule-level saturation curves.
04	04.DE_subsampling.R	Pseudobulk DE at each depth, then cluster HBDN501-AML-KMT2A and score separation.
05	05.LK1_resequenced_report.Rmd	This report.

Each analysis appears once. Where a script went through successive revisions, the version kept is the one that produced the outputs shown, and its header names the revisions it supersedes:

Step	Kept	Superseded
03	1c1... (2026-06-01)	1c... (2026-05-27, first pass). A later 1c2... (2026-06-16) also exists but produced no output - nothing under depth_benchmarking* postdates 2026-06-01 - and is shorter than the kept version, so it is not a strict superset.
04	1d2... (2026-06-01)	1d... (2026-05-28, first pass) and 1d3...DE_subsampling_LK2.R, which is byte-identical despite the _LK2 name and analyses LK1.

7. Tools

Tool	Version	Source
Cell Ranger	10.0.0	~/tools/cellranger-10.0.0/cellranger - the re-sequencing needed this version
Cell Ranger annotation	refdata-gex-GRCh38-2024-A	annotation/
AML scAtlas	published reference	results/AML_atlas/

```
## R version 4.5.1 (2025-06-13)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.5.2
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK version 3.12.1
##
## locale:
## [1] en_AU/en_AU/en_AU/C/en_AU/en_AU
##
## time zone: Australia/Melbourne
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices utils      datasets  methods   base
##
## other attached packages:
## [1] scales_1.4.0  ggplot2_4.0.2 knitr_1.51    purrr_1.2.1  stringr_1.6.0
## [6] readr_2.1.6   tidyr_1.3.2   dplyr_1.2.0
##
## loaded via a namespace (and not attached):
## [1] bit_4.6.0      gtable_0.3.6   jsonlite_2.0.0  crayon_1.5.3
## [5] qpdf_1.4.1     compiler_4.5.1 pdftools_3.9.0  Rcpp_1.1.1
## [9] tidyselect_1.2.1 parallel_4.5.1 jquerylib_0.1.4  yaml_2.3.12
## [13] fastmap_1.2.0  R6_2.6.1       generics_0.1.4  tibble_3.3.1
## [17] RColorBrewer_1.1-3 bslib_0.10.0   pillar_1.11.1   tzdb_0.5.0
## [21] rlang_1.1.7    cachem_1.1.0   stringi_1.8.7   xfun_0.56
## [25] S7_0.2.1       sass_0.4.10    bit64_4.6.0-1   otel_0.2.0
## [29] cli_3.6.5      withr_3.0.2    magrittr_2.0.4  digest_0.6.39
## [33] grid_4.5.1     vroom_1.7.0    askpass_1.2.1   hms_1.1.4
## [37] lifecycle_1.0.5 vctrs_0.7.1    evaluate_1.0.5  glue_1.8.0
## [41] farver_2.1.2   rmarkdown_2.30 tools_4.5.1     pkgconfig_2.0.3
## [45] htmltools_0.5.9
```

8. Caveats

1. The depth series here is Cell Ranger's molecule-level downsampling, not a FASTQ-level re-run. The seqtk FASTQ downsampling with `frac_000` to `frac_090` directories is a separate, later exercise carried out on LK2 (`cellranger_downsampled_reseq/260528...`, 2026-06-15) and does not belong to this report.
2. **Step 05 hardcodes a downsampled path under this run** that does not exist (`cellranger_downsampled_reseq/260522...`); only the LK2 equivalent is on disk. It runs against the molecule-level outputs regardless, but the path should be corrected before the script is reused.
3. The cluster-count table in section 4 is taken from the read-depth deck; the figure behind it is not among the saved outputs.